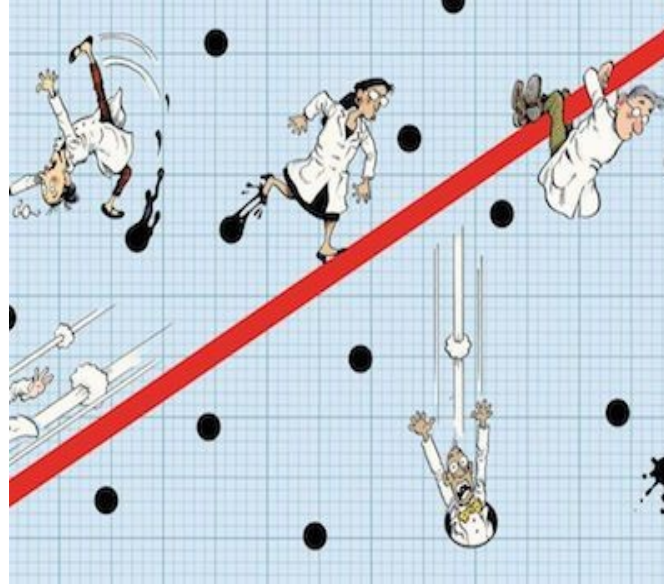


Introduction to statistical concepts & methods



Picture: Nature Publishing Group

open microbiome data science frameworks



QIIME 2™ is a next-generation microbiome bioinformatics platform that is extensible, free, open source, and [community developed](#).

[Code of Conduct »](#) [Citing QIIME 2 »](#) [Learn more »](#)



Automatically track your analyses with decentralized data provenance — no more guesswork on what commands were run!



Interactively explore your data with beautiful visualizations that provide new perspectives.



Easily share results with your team, even those members without QIIME 2 installed.



Plugin-based system — your favorite microbiome methods all in one place.



PeerJ >

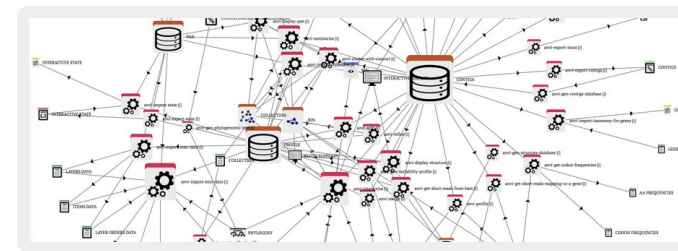
Anvi'o: an advanced analysis and visualization platform for 'omics data

[Research article](#) [Bioinformatics](#) [Biotechnology](#) [Computational Biology](#) [Genomics](#) [Microbiology](#)

A. Murat Eren^{1,2}, Özcan C. Esen¹, Christopher Quince³, Joseph H. Vineis¹, Hilary G. Morrison¹, Mitchell L. Sogin¹, Tom O. Delmont¹

Published October 8, 2015

Anvi'o in a nutshell



Anvi'o is an [open-source](#), [community-driven](#) analysis and visualization platform for 'omics data.

898,944

VIEWS

1,119

CITATIONS

4,143

SAVES

7,007

SHARES

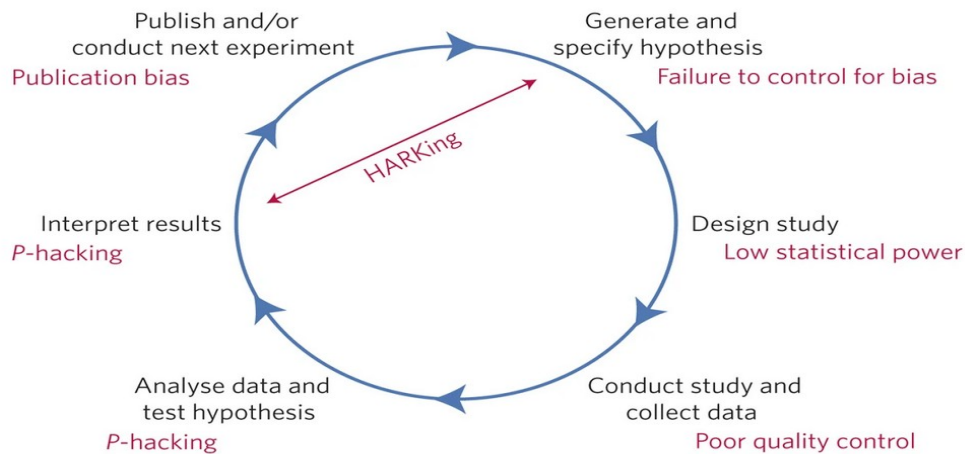
Why Most Published Research Findings Are False

John P. A. Ioannidis

Published: Aug 30, 2005 • DOI: [10.1371/journal.pmed.0020124](https://doi.org/10.1371/journal.pmed.0020124)

Figure 1: Threats to reproducible science.

From: [A manifesto for reproducible science](#)



An idealized version of the hypothetico-deductive model of the scientific method is shown. Various potential threats to this model exist (indicated in red), including lack of replication⁵, hypothesizing after the results are known (HARKing)⁷, poor study design, low statistical power², analytical flexibility⁵¹, P-hacking⁴, publication bias³ and lack of data sharing⁶. Together these will serve to undermine the robustness of published research, and may also impact on the ability of science to self-correct.

A manifesto for reproducible science

Marcus R. Munafò [✉](#), Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware & John P. A. Ioannidis

Nature Human Behaviour 1, Article number: 0021 (2017) | [Cite this article](#)

204k Accesses | 963 Citations | 2579 Altmetric | [Metrics](#)

Learning goals - Statistical concepts

- Overview of common pitfalls in statistical analyses
- Typical ways to deal with these

Contents

1) **All models are wrong, some are useful**

- Modeling assumptions, parametric vs. non-parametric models
- Supervised vs. unsupervised learning
- Statistical power

2) **Quantifying uncertainty**

- Bias, confounders, Simpson's effect
- Noise & errors
- Sensitivity & specificity, precision / recall, accuracy, FPR, TPR..

3) **Common pitfalls**

- Garden of the forking paths
- P-hacking
- Overfitting

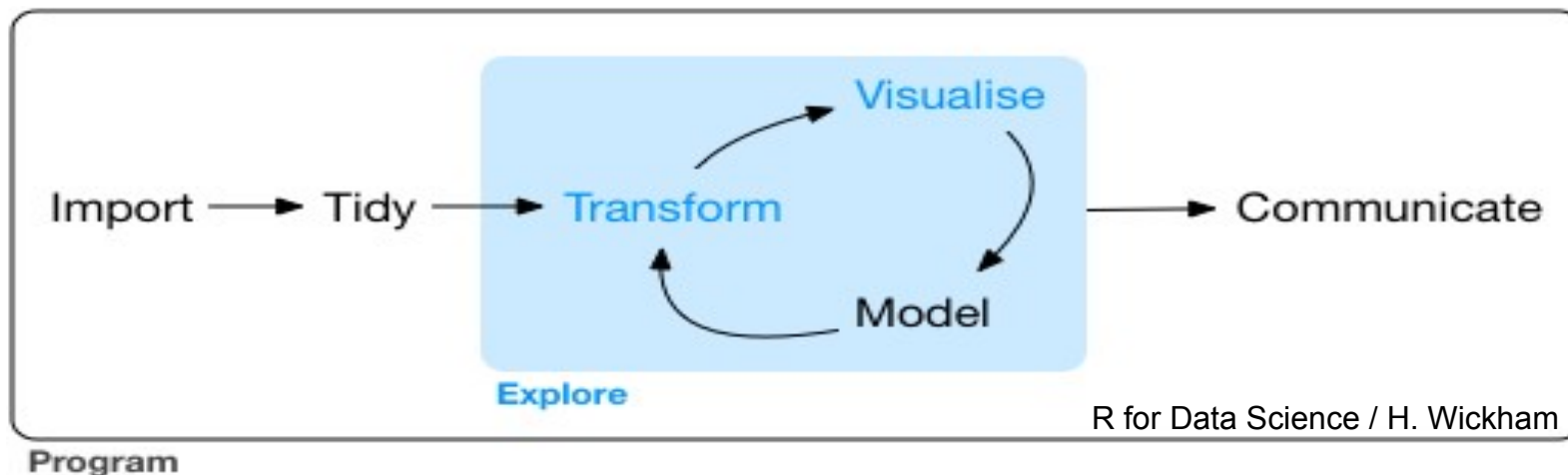
The influence of hidden researcher decisions in applied microeconomics

Nick Huntington-Klein ✉, Andreu Arenas, Emily Beam, Marco Bertoni, Jeffrey R. Bloem, Pralhad Burli, Naibin Chen, Paul Grieco, Godwin Ekpe, Todd Pugatch, Martin Saavedra, Yaniv Stopnitzky

First published: 22 March 2021

<https://doi.org/10.1111/ecin.12992>

Researchers make hundreds of decisions about data collection, preparation, and analysis in their research. We use a many-analysts approach to measure the extent and impact of these decisions. Two published causal empirical results are replicated by seven replicators each. We find large differences in data preparation and analysis decisions, many of which would not likely be reported in a publication. No two replicators reported the same sample size. Statistical significance varied across replications, and for one of the studies the effect's sign varied as well. The standard deviation of estimates across replications was 3–4 times the mean reported standard error.



Fundamental considerations in beta diversity analysis

Feature selection

(all/core taxa; genus/strain level..?)

Transformation

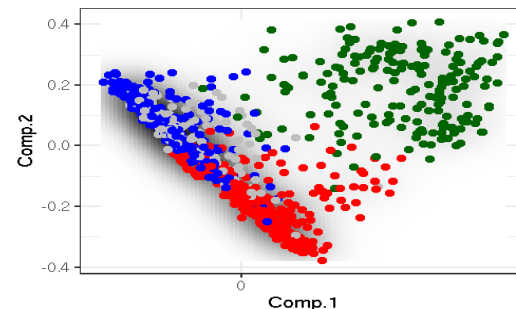
(absolute, compositional, CLR, Hellinger..?)

Dissimilarity measure

(Euclidean/L2, Bray-Curtis, Unifrac..?)

Analysis method

(PCA, PCoA, NMDS, t-SNE, UMAP..)



How to choose a correct model?

→ a community typing example

Taxonomic level

- Phylum
- Family
- Order
- Genus
- Species
- Strain..

Filtering

- None
- Prevalent
- Core
- Excl. outliers
- High variance
- Custom

Normalization

- None
- TSS
- CSS
- ILR/ALR/CLR
- phILR
- Hellinger

(Dis)similarity

- Eulidean
- Aitchison
- Bray-Curtis
- Jaccard
- weighted Unifrac
- unweighted Unifrac

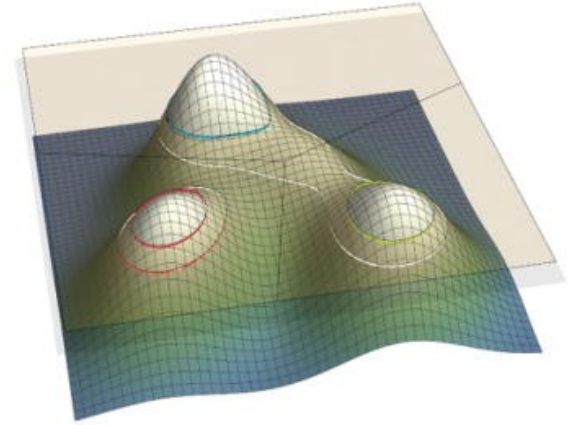
Clustering method

- Hierarchical / Ward
- Hierarchical / Complete
- Gaussian mixture
- DMM
- PAMR
- K-means

Regulation

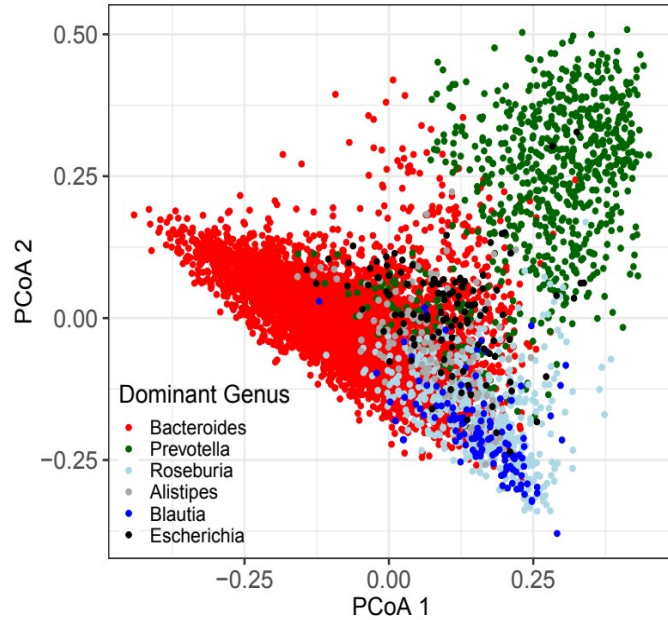
- Calinski-Harabasz
- Dirichlet Process
- Silhouette Index
- AIC
- BIC
- DIC

$$2 \times 6^6 = 93312$$

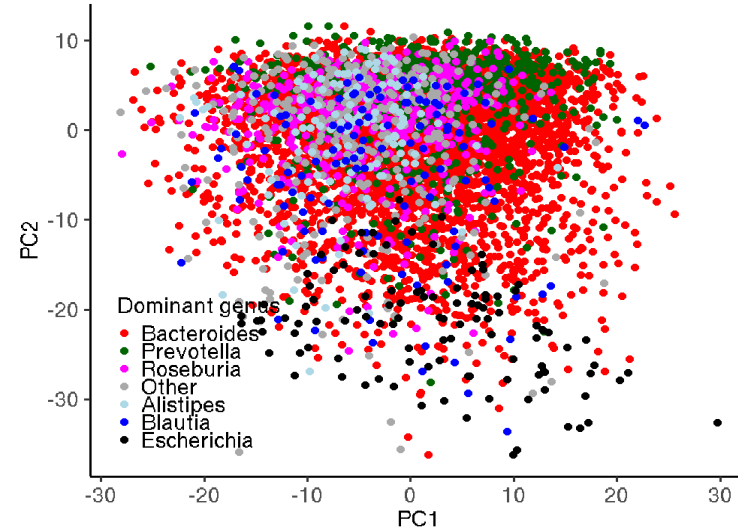


Enterotypes in the landscape of gut microbial community composition. Costea *et al.* Nature 2018.

PCoA + Bray-Curtis



PCA + Aitchison



Reproducible Research: Enterotype Example

Susan Holmes and Joey McMurdie

<http://statweb.stanford.edu/~susan/papers/EnterotypeRR.html>

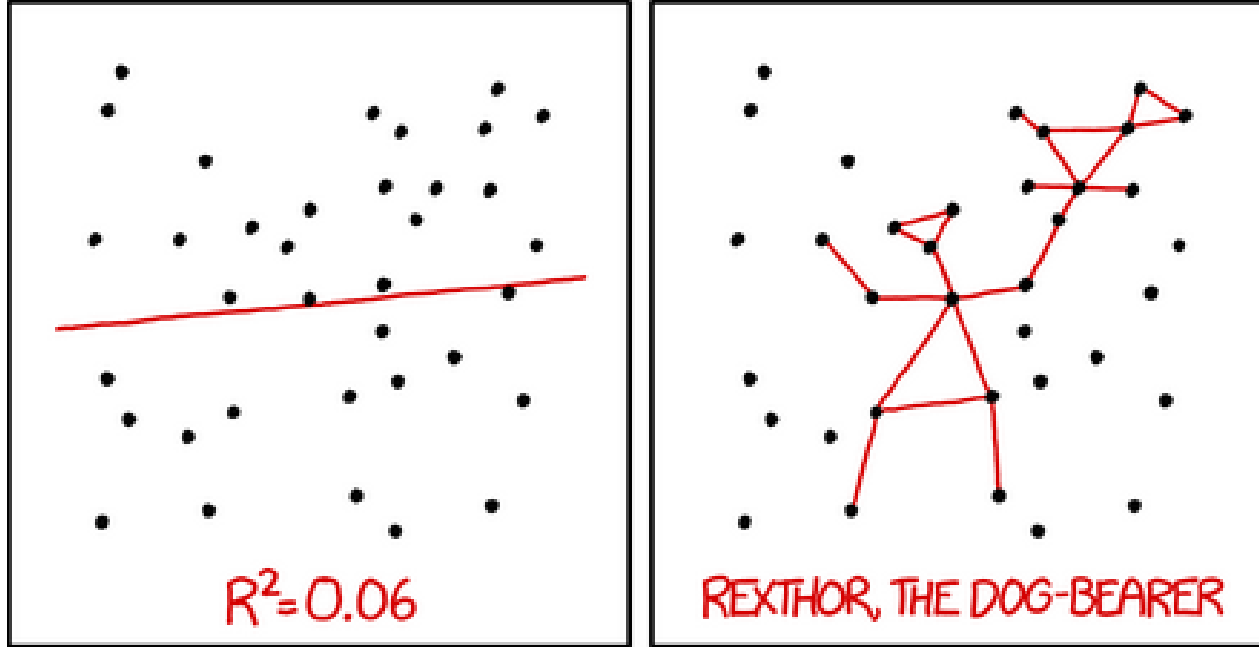
[Comment on this paper](#)

Taxonomic Signatures of Long-Term Mortality Risk in Human Gut Microbiota

Aaro Salosensaari, Ville Laitinen, Aki Havulinna, Guillaume Meric, Susan Cheng, Markus Perola, Liisa Valsta, Georg Alfthan, Michael Inouye, Jeramie D. Watrous, Tao Long, Rodolfo Salido, Karenina Sanders, Caitriona Brennan, Gregory C. Humphrey, Jon G. Sanders, Mohit Jain, Pekka Jousilahti, Veikko Salomaa, Rob Knight, Leo Lahti, Teemu Niiranen

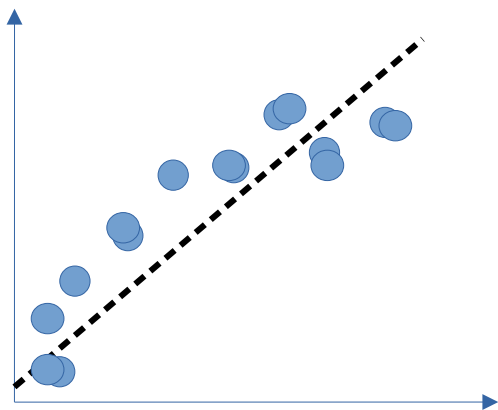
doi: <https://doi.org/10.1101/2019.12.30.19015842>

How we choose which model to apply?

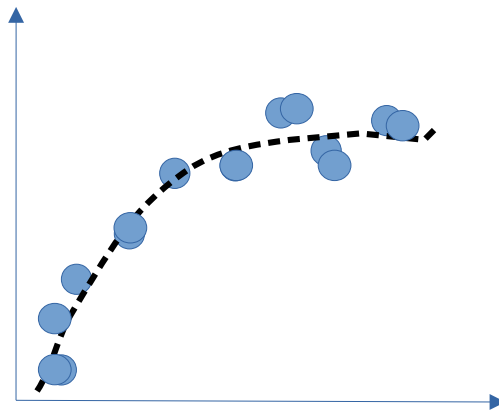


I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER
TO GUESS THE DIRECTION OF THE CORRELATION FROM THE
SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

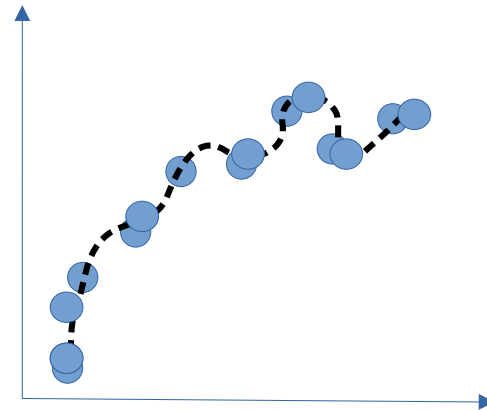
Tradeoff between robustness and sensitivity



Simple model:
underfitting &
high bias



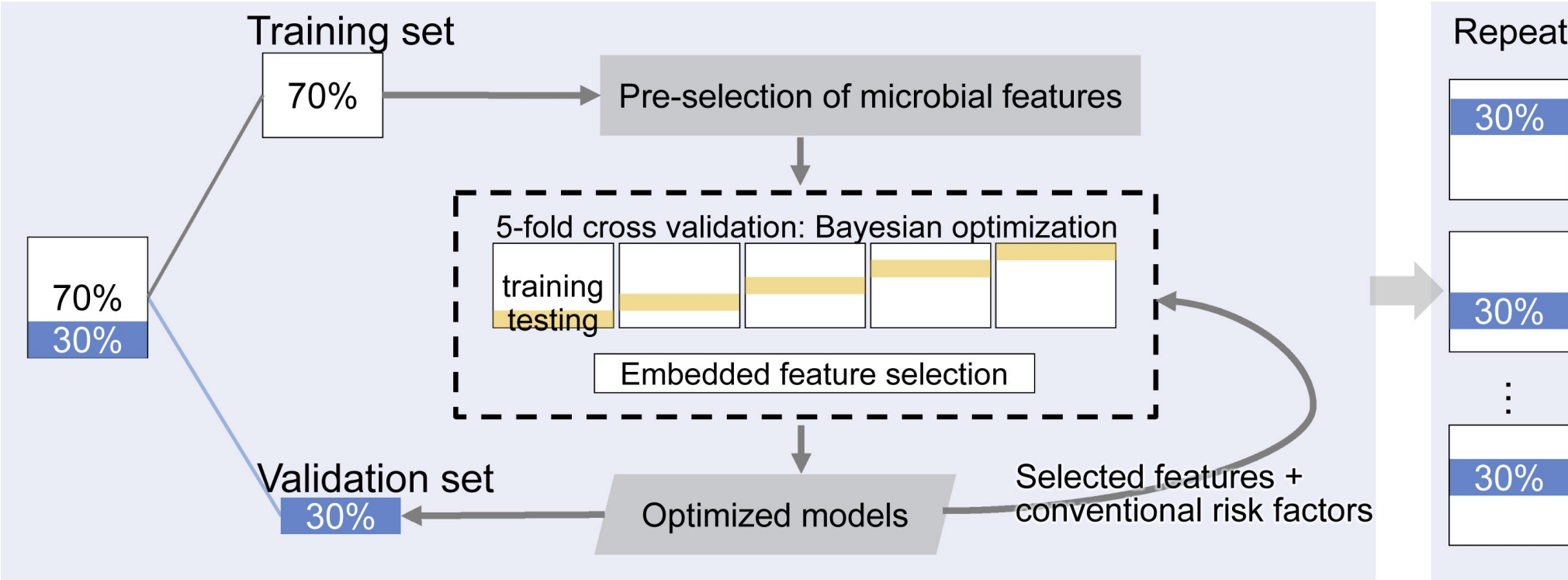
Intermediate
model –
"just right"



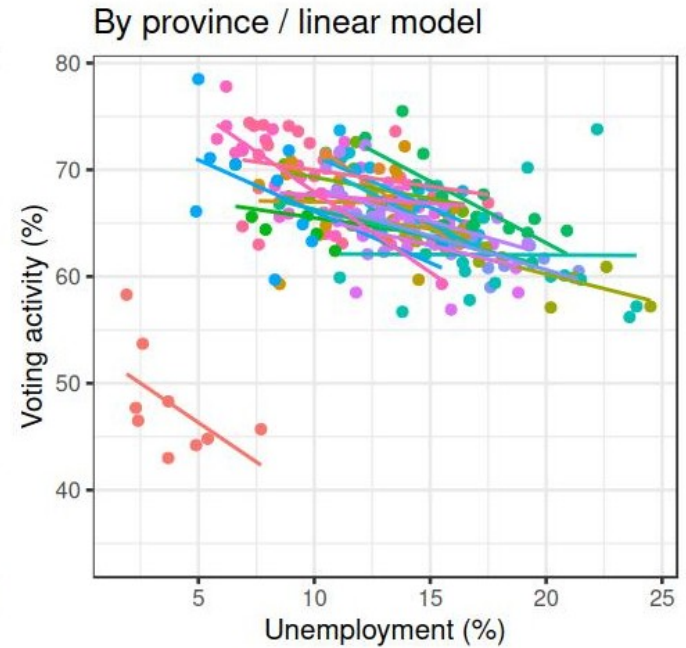
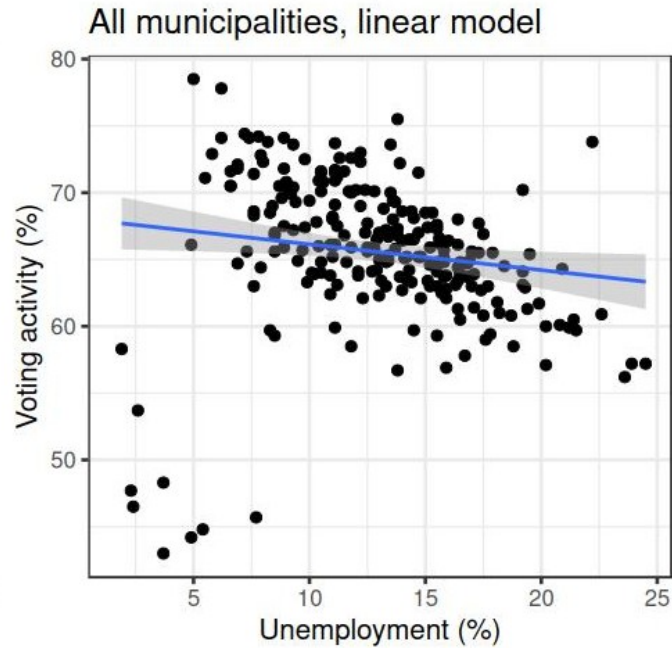
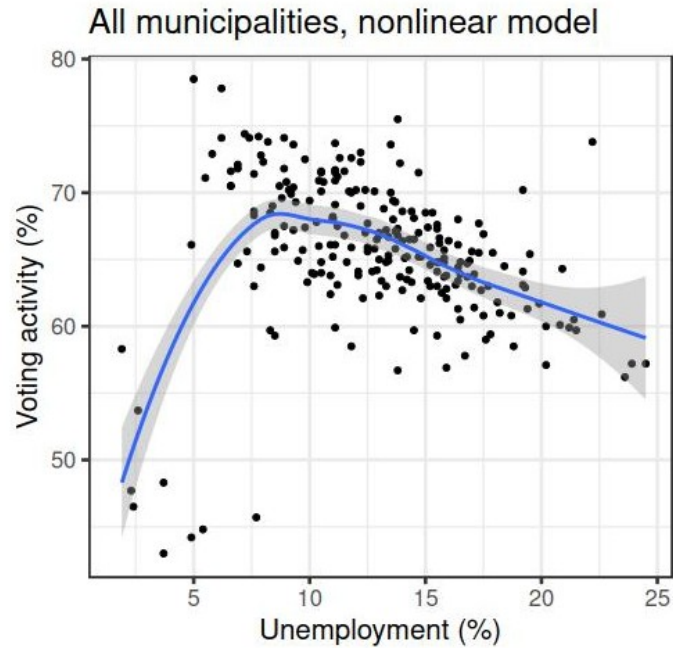
Complex model –
overfitting &
high variance

Early prediction of incident liver disease using conventional risk factors and gut-microbiome-augmented gradient boosting

[Yang Liu](#) ^{1,2,24} [✉](#) · [Guillaume Méric](#) ^{1,2,3,4} · [Aki S. Havulinna](#) ^{5,6} · ... · [Veikko Salomaa](#) ⁵ · [Rob Knight](#) ^{12,17,18} · [Michael Inouye](#) ^{1,2,3,7,20,21,22,23} [✉](#) ... [Show more](#)



Simpson's effect

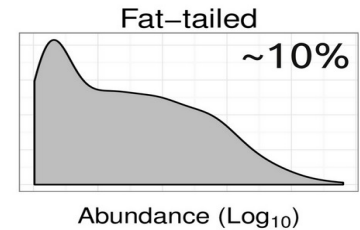
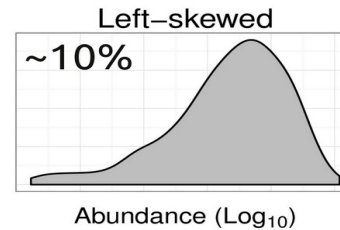
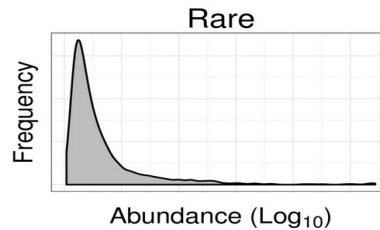
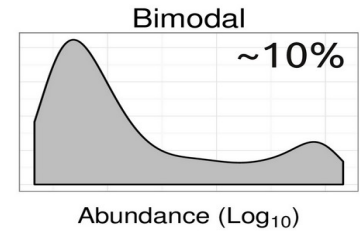
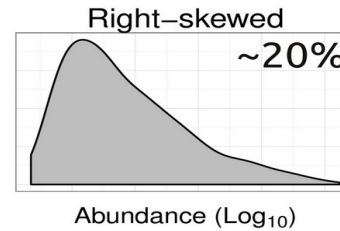
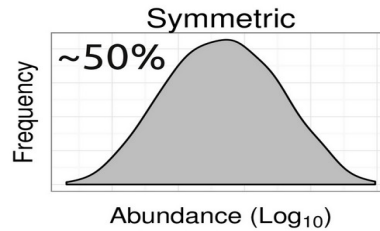


Differential abundance

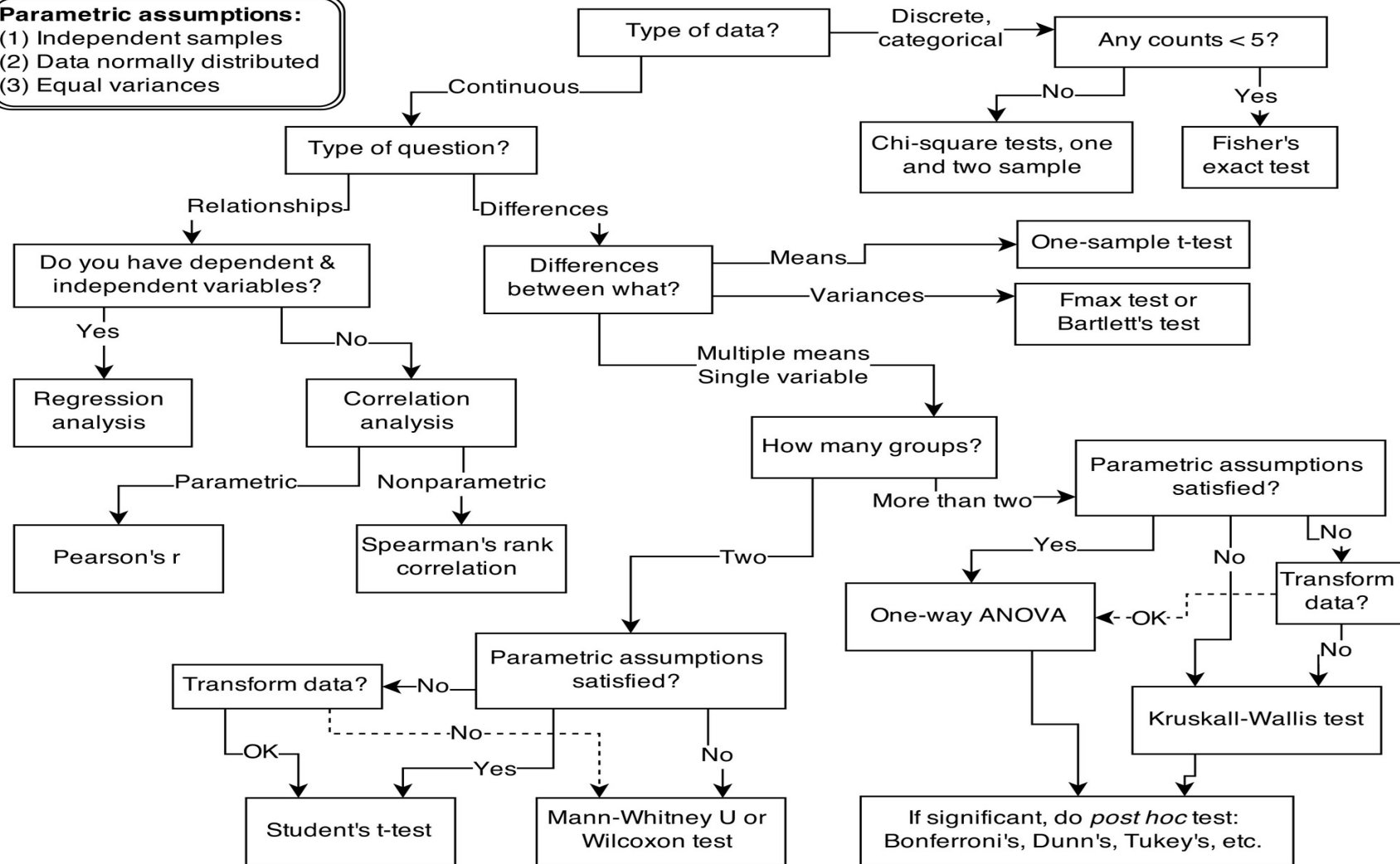
Standard t-test for two-group comparison?

Problems:

- Few replicates
- Non-gaussian, discrete, positive, skewed..
- Multiple testing

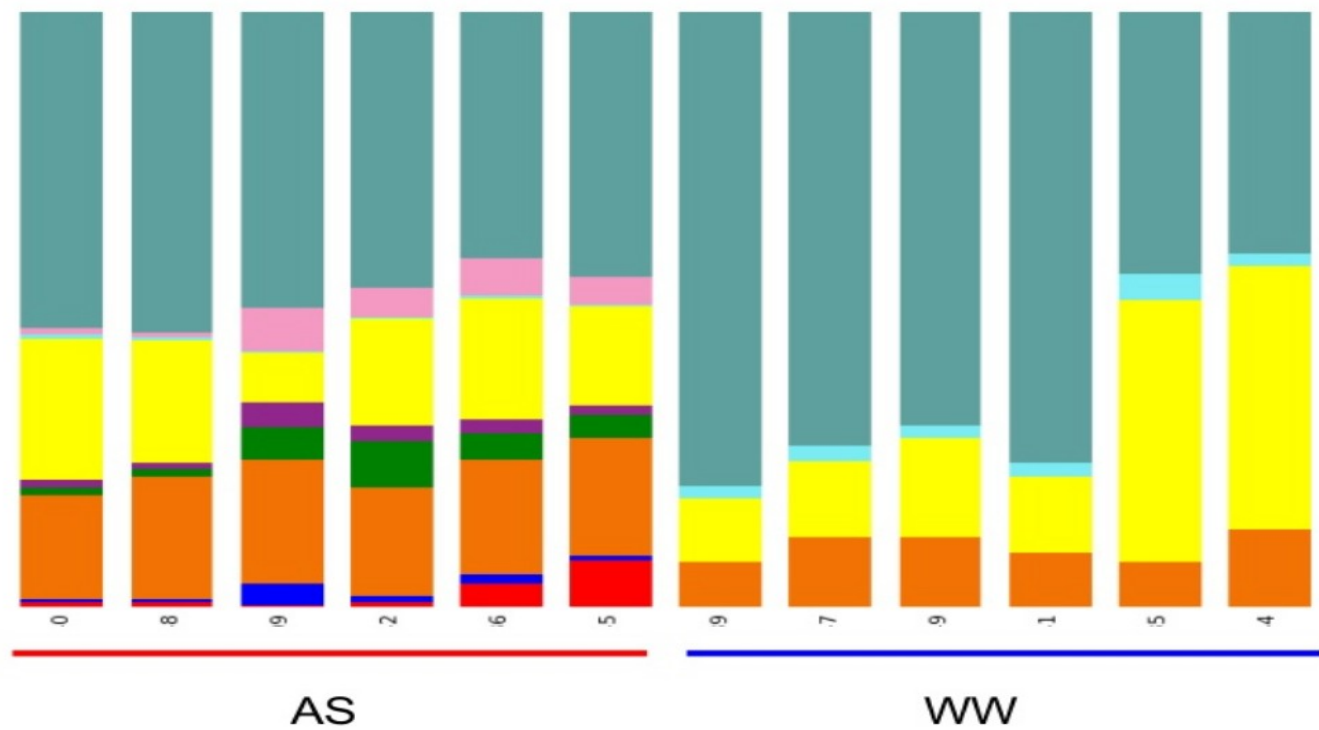


Parametric assumptions:
 (1) Independent samples
 (2) Data normally distributed
 (3) Equal variances

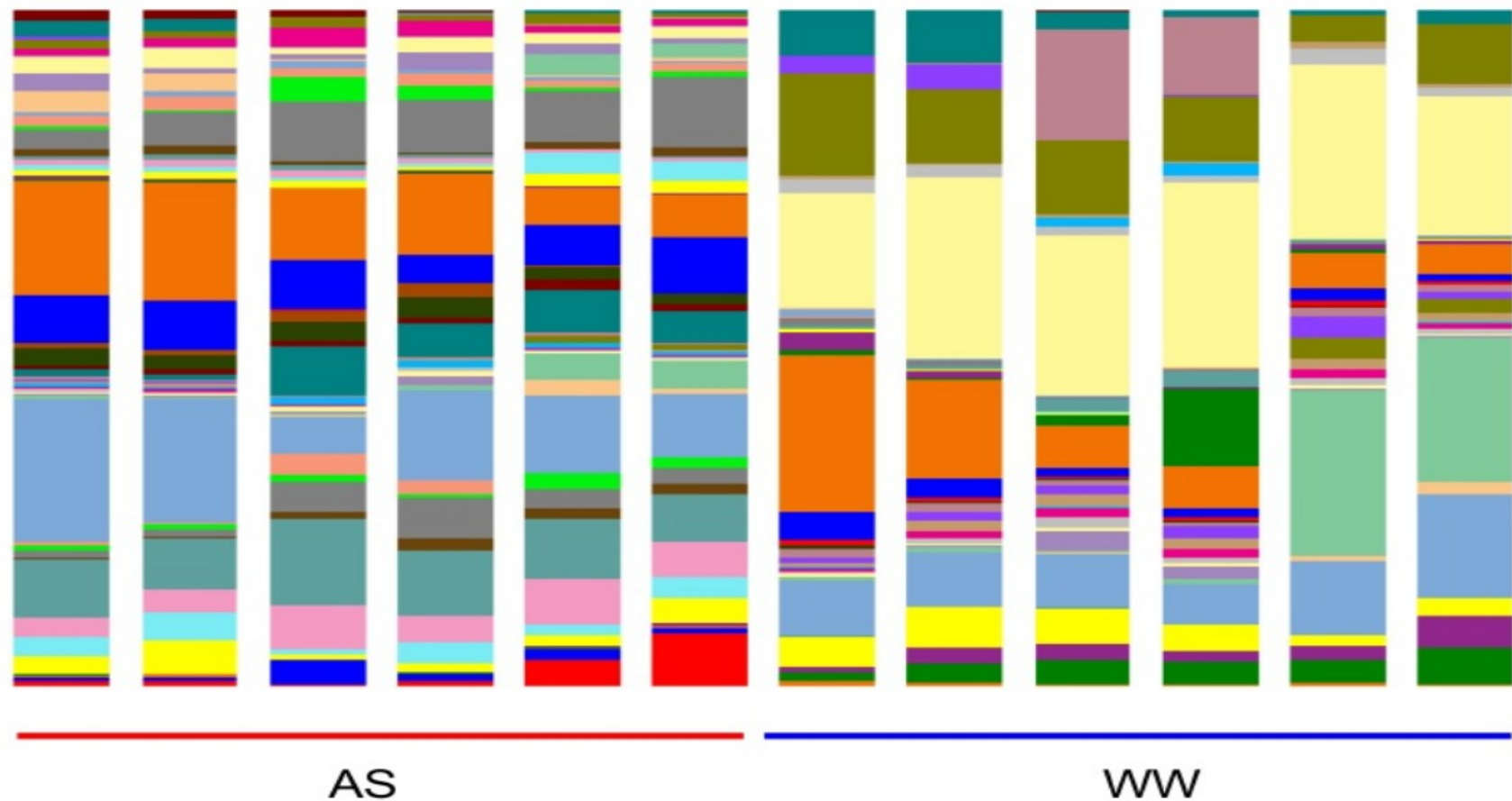


(barely) not statistically significant (p=0.052)	an expected trend (p=0.08)	at the cusp of significance (p=0.06)	effectively significant (p=0.051)	just very slightly missed the significance level (p=0.086)
a barely detectable statistically significant difference (p=0.073)	an important trend (p=0.066)	at the edge of significance (p=0.055)	equivocal significance (p=0.06)	leaning towards significance (p=0.15)
a borderline significant trend (p=0.09)	an increasing trend (p<0.09)	at the limit of significance (p=0.054)	essentially significant (p=0.10)	leaning towards statistical significance (p=0.06)
a certain trend toward significance (p=0.08)	an interesting trend (p=0.1)	at the limits of significance (p=0.053)	extremely close to significance (p=0.07)	likely to be significant (p=0.054)
a clear tendency to significance (p=0.052)	an inverse trend toward significance (p=0.06)	at the margin of significance (p=0.056)	failed to reach significance on this occasion (p=0.09)	loosely significant (p=0.10)
a clear trend (p<0.09)	an observed trend (p=0.06)	at the margin of statistical significance (p<0.07)	failed to reach statistical significance (p=0.06)	marginal significance (p=0.07)
a clear, strong trend (p=0.09)	an obvious trend (p=0.06)	at the verge of significance (p=0.058)	fairly close to significance (p=0.065)	marginally and negatively significant (p=0.08)
a considerable trend toward significance (p=0.069)	an overall trend (p=0.2)	at the very edge of significance (p=0.053)	fairly significant (p=0.09)	marginally insignificant (p=0.08)
a decreasing trend (p=0.09)	an unexpected trend (p=0.09)	barely below the level of significance (p=0.06)	falls just short of standard levels of statistical significance (p=0.06)	marginally nonsignificant (p=0.096)
a definite trend (p=0.08)	an unexplained trend (p=0.09)	barely escaped statistical significance (p=0.07)	fell (just) short of significance (p=0.08)	marginally outside the level of significance
a distinct trend toward significance (p=0.07)	an unfavorable trend (p<0.10)	barely escapes being statistically significant at the 5% risk level (0.1>p>0.05)	fell barely short of significance (p=0.08)	marginally significant (p>0.1)
a favorable trend (p=0.09)	appeared to be marginally significant (p<0.10)	barely failed to attain statistical significance (p=0.067)	fell just short of significance (p=0.07)	marginally significant tendency (p=0.08)
a favourable statistical trend (p=0.09)	approached acceptable levels of statistical significance (p=0.054)	barely fails to attain statistical significance at conventional levels (p<0.10)	fell just short of statistical significance (p=0.12)	marginally statistically significant (p=0.08)
a little significant (p<0.1)	approached but did not quite achieve significance (p>0.05)	barely insignificant (p=0.075)	fell just short of the traditional definition of statistical significance (p=0.051)	may not be significant (p=0.06)
a margin at the edge of significance (p=0.0608)	approached but fell short of significance (p=0.07)	barely missed statistical significance (p=0.051)	fell marginally short of significance (p=0.07)	medium level of significance (p=0.051)
a marginal trend (p=0.09)	approached conventional levels of significance (p<0.10)	barely missed the commonly acceptable significance level (p<0.053)	fell narrowly short of significance (p=0.0623)	mildly significant (p=0.07)
a marginal trend toward significance (p=0.052)	approached near significance (p=0.06)	barely outside the range of significance (p=0.06)	fell only marginally short of significance (p=0.0879)	missed narrowly statistical significance (p=0.054)
a marked trend (p=0.07)	approached our criterion of significance (p>0.08)	barely significant (p=0.07)	fell only short of significance (p=0.06)	moderately significant (p>0.11)
a mild trend (p<0.09)	approached the borderline of significance (p=0.07)	below (but verging on) the statistical significant level (p>0.05)	fell short of significance (p=0.07)	modestly significant (p=0.09)
a moderate trend toward significance (p=0.068)	approached the level of significance (p=0.09)	better trends of improvement (p=0.056)	fell slightly short of significance (p>0.0167)	narrowly avoided significance (p=0.052)
a near-significant trend (p=0.07)	approached trend levels of significance (p0.05)	bordered on a statistically significant value (p=0.06)	fell somewhat short of significance (p=0.138)	narrowly eluded statistical significance (p=0.0789)
a negative trend (p=0.09)	approached, but did reach, significance (p=0.065)	bordered on being significant (p>0.07)	felt short of significance (p=0.07)	narrowly escaped significance (p=0.08)
a nonsignificant trend (p<0.1)	approaches but fails to achieve a customary level of statistical significance (p=0.154)	bordered on being statistically significant (p=0.0502)	flirting with conventional levels of significance (p>0.1)	narrowly evaded statistical significance (p>0.05)
a nonsignificant trend toward significance (p=0.1)	approaches statistical significance (p>0.06)	bordered on but was not less than the accepted level of significance (p>0.05)	heading towards significance (p=0.086)	narrowly failed significance (p=0.054)
a notable trend (p<0.1)	approaching a level of significance (p=0.089)	bordered on significant (p=0.09)	highly significant (p=0.09)	narrowly missed achieving significance (p=0.055)
a numerical increasing trend (p=0.09)	approaching an acceptable significance level (p=0.056)	borderline conventional significance (p=0.051)	hint of significance (p>0.05)	narrowly missed overall significance (p=0.06)
a numerical trend (p=0.09)	approaching borderline statistical significance (p=0.07)	borderline level of statistical significance (p=0.053)	hovered around significance (p = 0.061)	narrowly missed significance (p=0.051)
a positive trend (p=0.09)	approaching close to significance (p<0.1)	borderline significant (p=0.09)	hovered at nearly a significant level (p=0.058)	narrowly missed standard significance levels (p<0.07)
a possible trend (p=0.09)	approaching conventional significance levels (p=0.06)	borderline significant trends (p=0.099)	hovering closer to statistical significance (p=0.076)	narrowly missed the significance level (p=0.07)
a possible trend toward significance (p=0.052)	approaching independent prognostic significance (p=0.08)	close to a marginally significant level (p=0.06)	hovers on the brink of significance (p=0.055)	narrowly missing conventional significance (p=0.054)
a pronounced trend (p=0.09)	approaching marginal levels of significance (p<0.107)	close to being significant (p=0.06)	in the edge of significance (p=0.059)	near limit significance (p=0.073)
a reliable trend (p=0.058)		close to being statistically significant (p=0.055)	in the verge of significance (p=0.06)	near miss of statistical significance (p>0.1)
a robust trend toward significance (p=0.0503)		close to borderline significance (p=0.072)	inconclusively significant (p=0.070)	near nominal significance (p=0.064)
a significant trend (p=0.09)		close to the boundary of significance (p=0.06)	indeterminate significance (p=0.08)	near significance (p=0.07)
a slight slide towards significance (p<0.20)		close to the level of significance (p=0.07)	indicative significance (p=0.08)	near to statistical significance (p=0.056)
a slight tendency toward significance(p<0.08)		close to the limit of significance (p=0.17)	is just outside the conventional levels of significance	near/possible significance(p=0.0661)
a slight trend (p<0.09)		close to the margin of significance (p=0.055)	just above significant (p=0.051)	near-borderline significance (p=0.10)
a slight trend toward significance (p=0.098)		close to the margin of statistical significance (p=0.075)	just above the arbitrary level of significance (p=0.07)	near-certain significance (p=0.07)
a slightly increasing trend (p=0.09)		closely approaches the brink of significance (p=0.07)	just above the margin of significance (p=0.053)	nearing significance (p<0.051)
a small trend (p=0.09)		closely approaches the statistical significance (p=0.0669)	just at the conventional level of significance (p=0.05001)	nearly acceptable level of significance (p=0.06)
a statistical trend (p=0.09)		closely approximating significance (p>0.05)	just barely below the level of significance (p=0.06)	nearly approaches statistical significance (p=0.079)
a statistical trend toward significance (p=0.09)		closely not significant (p=0.06)	just barely failed to reach significance (p<0.06)	nearly borderline significance (p=0.052)
a strong tendency towards statistical significance (p=0.051)		closely significant (p=0.058)	just barely insignificant (p=0.11)	nearly negatively significant (p<0.1)
a strong trend (p=0.077)		close-to-significant (p=0.09)	just barely statistically significant (p=0.054)	nearly positively significant (p=0.063)
a strong trend toward significance (p=0.08)		close to achieve conventional threshold levels of statistical significance (p=0.08)	just beyond significance (p=0.06)	nearly reached a significant level (p=0.07)
a substantial trend toward significance (p=0.068)		did not achieve conventional level of statistical significance (p<0.08)	just borderline significant (p=0.058)	nearly reaching the level of significance (p<0.06)
a suggestive trend (p=0.06)		did not quite achieve acceptable levels of statistical significance (p=0.054)	just escaped significance (p=0.07)	nearly significant (p=0.06)
a trend close to significance (p=0.08)		did not quite achieve significance (p=0.076)	just failed significance (p=0.057)	nearly significant tendency (p=0.06)
a trend significance level (p=0.08)		did not quite achieve the conventional levels of significance (p=0.052)	just failed to be significant (p=0.072)	nearly, but not quite significant (p=0.06)
a trend that approached significance (p<0.06)			just failed to reach statistical significance (p=0.06)	near-marginal significance (p=0.18)
a very slight trend toward significance (p=0.20)			just failing to reach statistical significance (p=0.06)	near-significant (p=0.09)
a weak trend (p=0.09)			just fails to reach conventional levels of statistical significance (p=0.07)	near-to-significance (p=0.093)
a weak trend toward significance (p=0.12)				near-trend significance (p=0.11)
an evident trend (p=0.13)				nominally significant (p=0.08)
				non-insignificant result (p=0.500)
				non-significant in the statistical sense (p>0.05)
				not absolutely significant but very probably so (p>0.05)
				not as significant (p=0.06)
				not clearly significant (p=0.08)
				not completely significant (p=0.07)

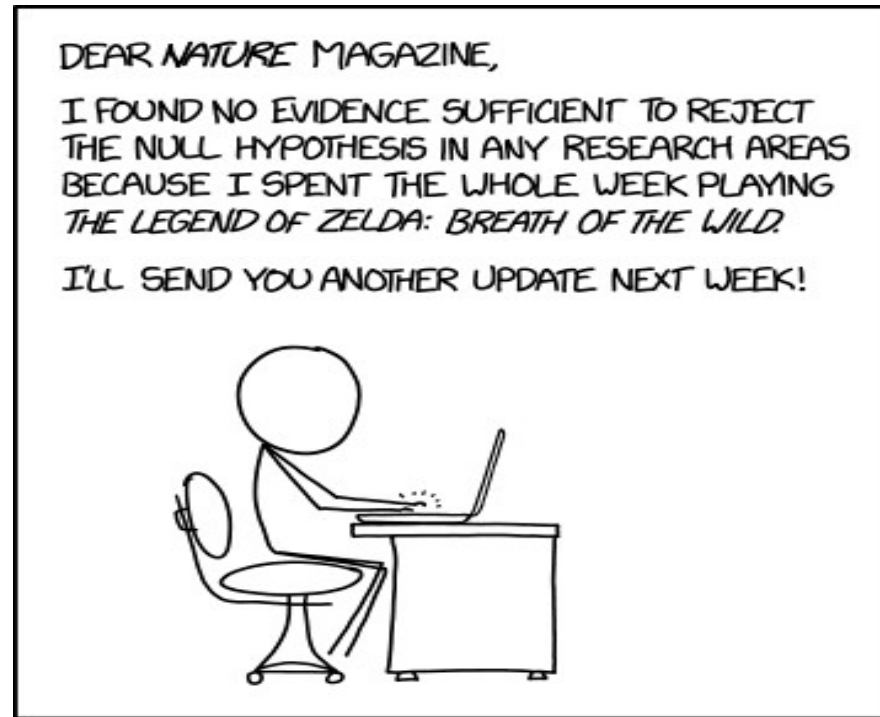
Phylum level Classification



Genus level Classification



*absence of evidence
is not
evidence of absence*



DEAR NATURE MAGAZINE,

I FOUND NO EVIDENCE SUFFICIENT TO REJECT
THE NULL HYPOTHESIS IN ANY RESEARCH AREAS
BECAUSE I SPENT THE WHOLE WEEK PLAYING
THE LEGEND OF ZELDA: BREATH OF THE WILD.

I'LL SEND YOU ANOTHER UPDATE NEXT WEEK!

THE PUSH TO PUBLISH NEGATIVE RESULTS SEEMS
KINDA WEIRD, BUT I'M HAPPY TO GO ALONG WITH IT.

Learning goals - Statistical methods

Overview of common methods for statistical analyses

1) Data cleaning

- Normalization, denoising & preprocessing

2) Data analysis

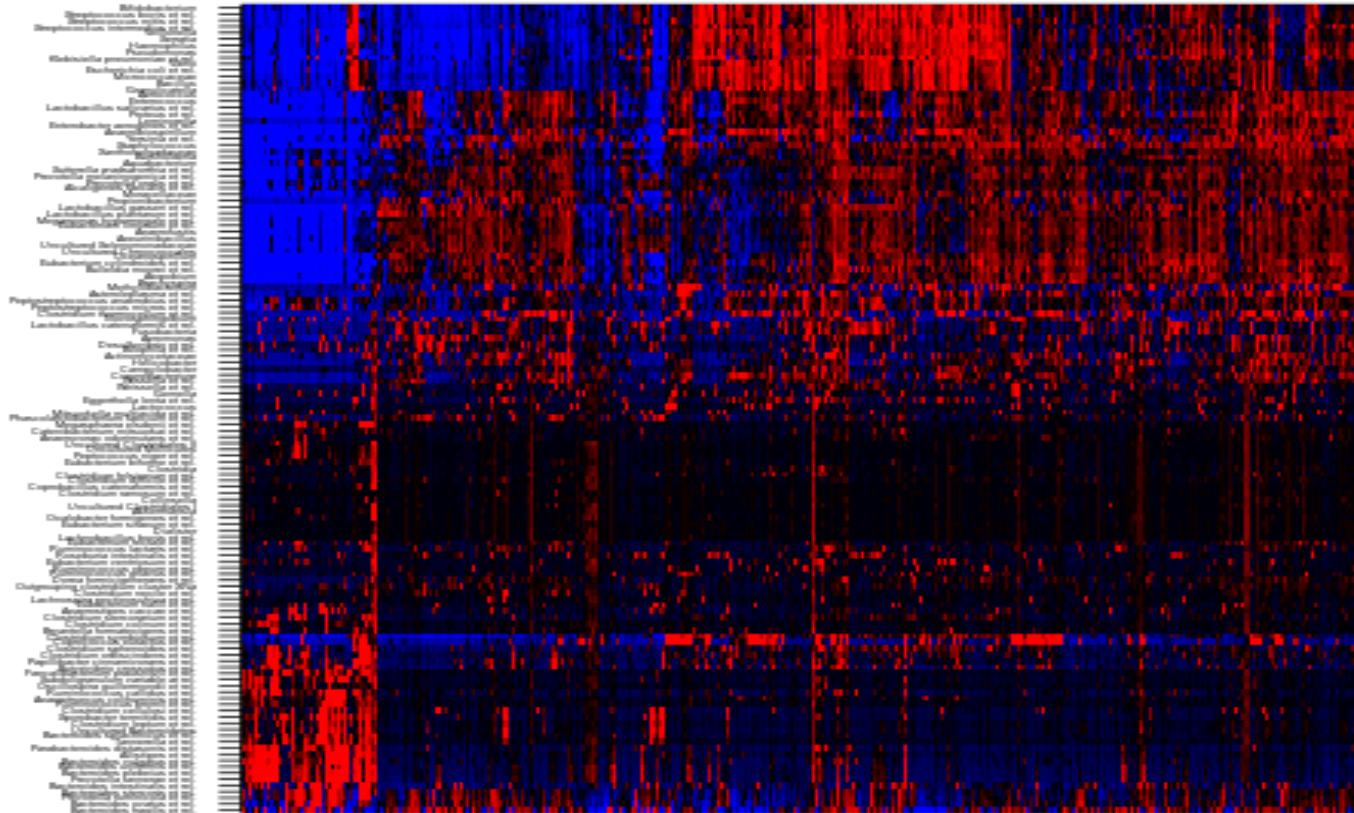
- Linear model vs. nonlinear models
- Parametric vs. non-parametric tests
- Bias & confounders: modeling the covariates
- Power calculations

3) Model validation

- Overfitting & cross-validation (internal, external)
- Garden of the forking paths: model comparisons & information criteria
- P-hacking: sensitivity analyses

4) Data visualization

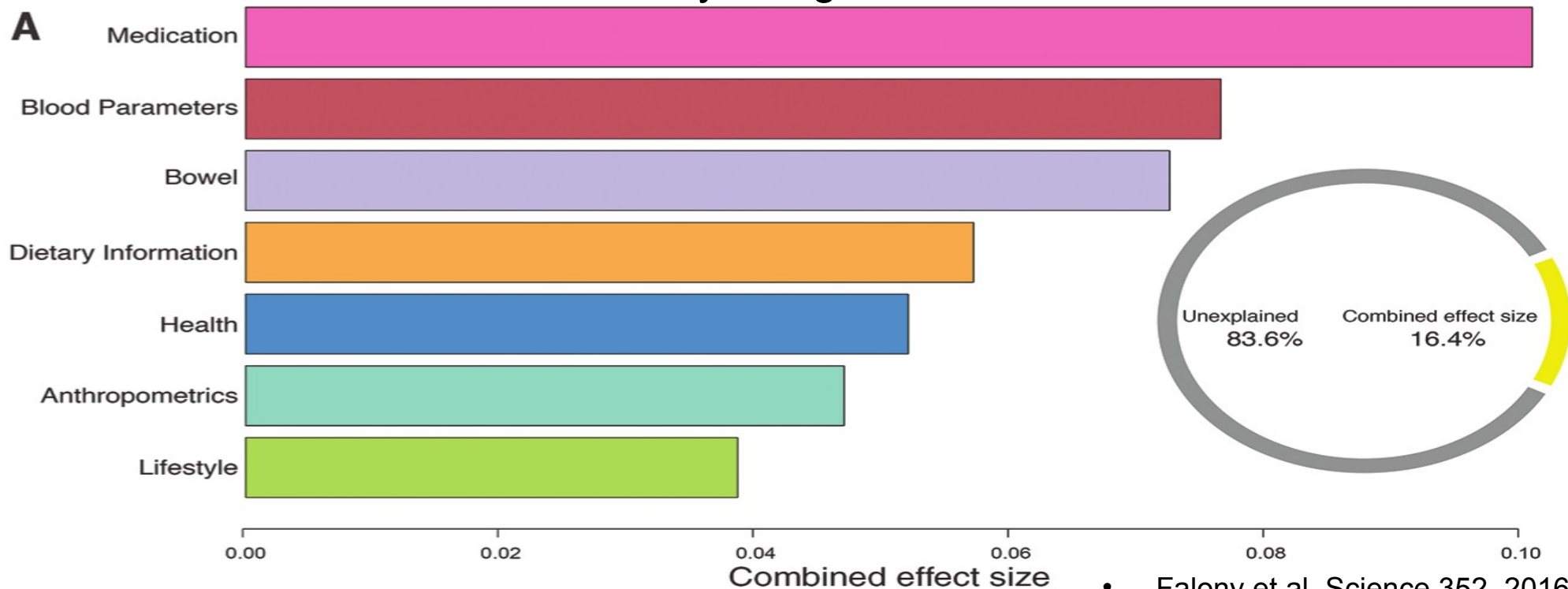
Organisms and samples are not independent understanding & modeling the (latent) structure(s)



Total explained variation: 16.4%

(Flemish Gut Flora Project)

- Proposed disease marker genera associated to host covariates and medication - inclusion in study design is essential !

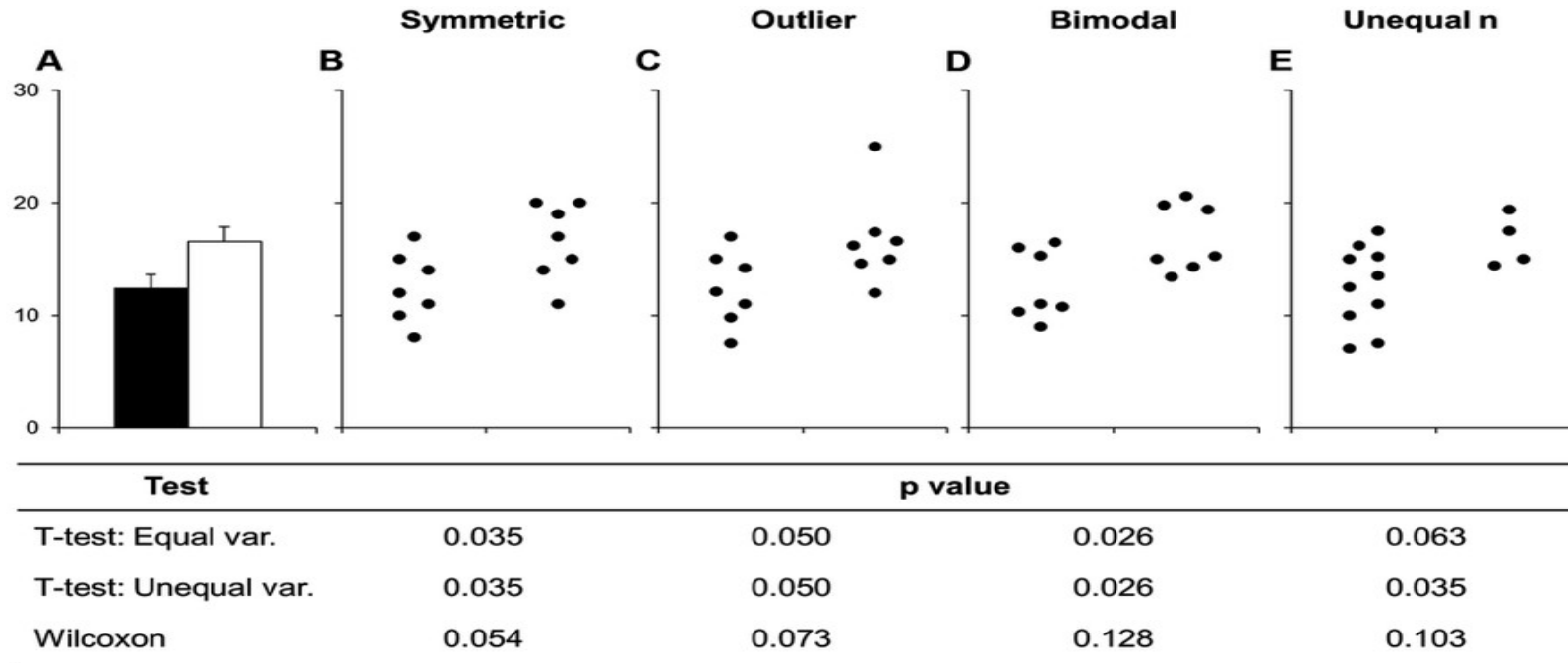


• Falony et al. Science 352, 2016.

$P < 0.04$
Effect?

$P < 0.05$

$P < 0.06$
No effect?



The ASA Statement on *p*-Values: Context, Process, and Purpose

Ronald L. Wasserstein & Nicole A. Lazar

Pages 129-133 | Accepted author version posted online: 07 Mar 2016, Published online: 09 Jun 2016

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*ASA advises researchers to avoid drawing scientific conclusions or making policy decisions based on *P* values alone. Researchers should describe not only the data analyses that produced statistically significant results, the society says, but all statistical tests and choices made in calculations. Otherwise, results may seem falsely robust. “the increased quantification of scientific research and a proliferation of large, complex data sets has expanded the scope for statistics and the importance of appropriately chosen techniques, properly conducted analyses, and correct interpretation”*

“statisticians often supplement or even replace *p*-values with other approaches. These include methods “that emphasize estimation over testing such as confidence, credibility, or prediction intervals; Bayesian methods; alternative measures of evidence such as likelihood ratios or Bayes factors; and other approaches such as decision-theoretic modeling and false discovery rates.”

- 1) *P*-values can indicate how incompatible the data are
- 2) with a specified statistical model.
- 3) *P*-values do not measure the probability that the studied
- 4) hypothesis is true, or the probability that the data were
- 5) produced by random chance alone.
- 6) Scientific conclusions and business or policy decisions
- 7) should not be based only on whether a *p*-value passes
- 8) a specific threshold.
- 9) Proper inference requires full reporting and transparency.
- 10) A *p*-value, or statistical significance, does not measure the
- 11) size of an effect or the importance of a result.
- 12) By itself, a *p*-value does not provide a good measure of
- 13) evidence regarding a model or hypothesis.

“The *p*-value was never intended to be a substitute for scientific reasoning,” said Ron Wasserstein, the ASA’s executive director. “Well-reasoned statistical arguments contain much more than the value of a single number and whether that number exceeds an arbitrary threshold. The ASA statement is intended to steer research into a ‘post $p < 0.05$ era.’”

Taylor's law (in HITChip Atlas)

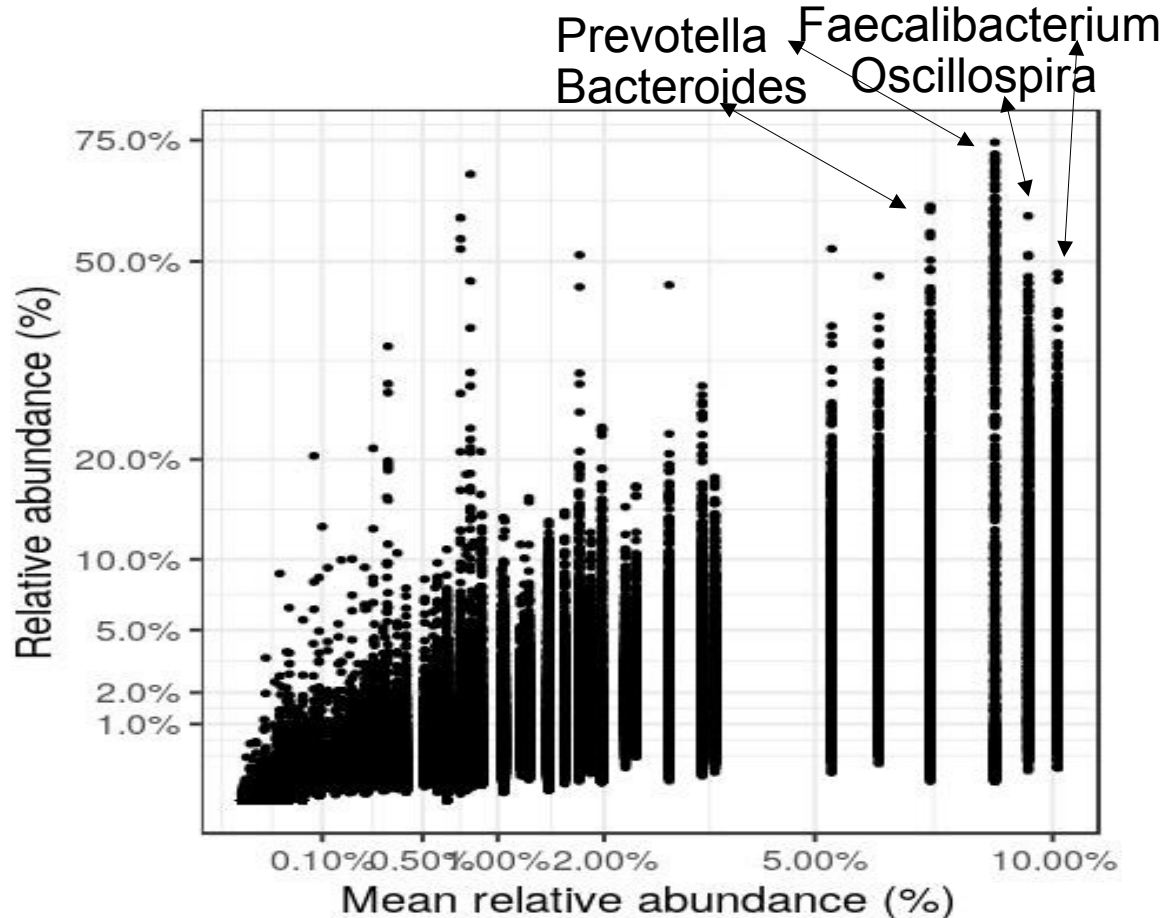
Heteroschedasticity:

Variance increases with the mean

Overdispersion:

Variance increases faster than proposed by the model

Data: HITChip Atlas

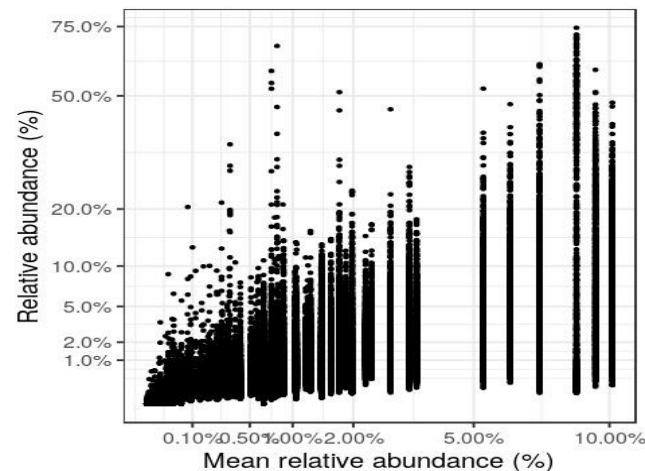


Microbiome data properties

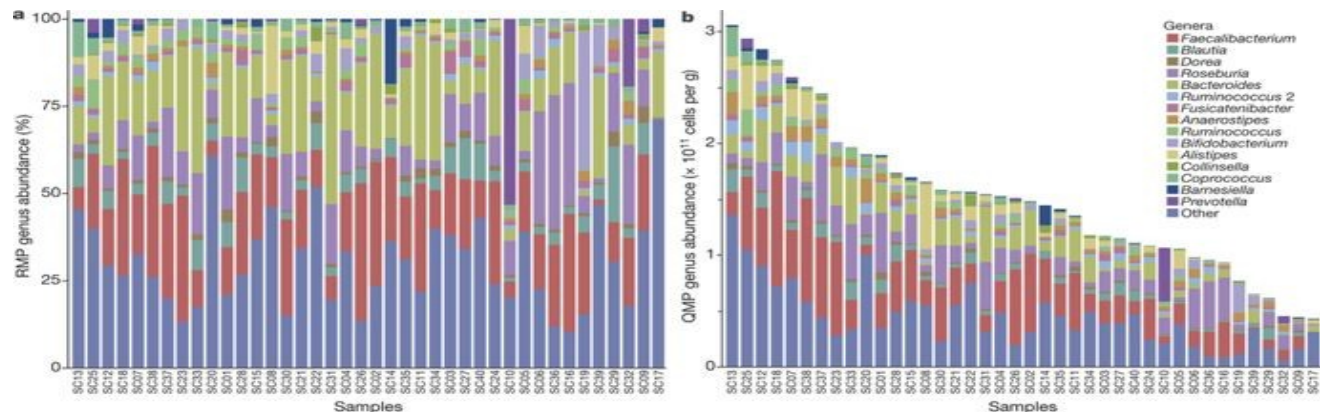
- Sparse
- Non-Gaussian
- Overdispersed
- Compositional
- Complex
- Stochastic
- Hierarchical

Heteroschedasticity

Data: HITChip Atlas (Lahti et al.
Nature Communications 2014)



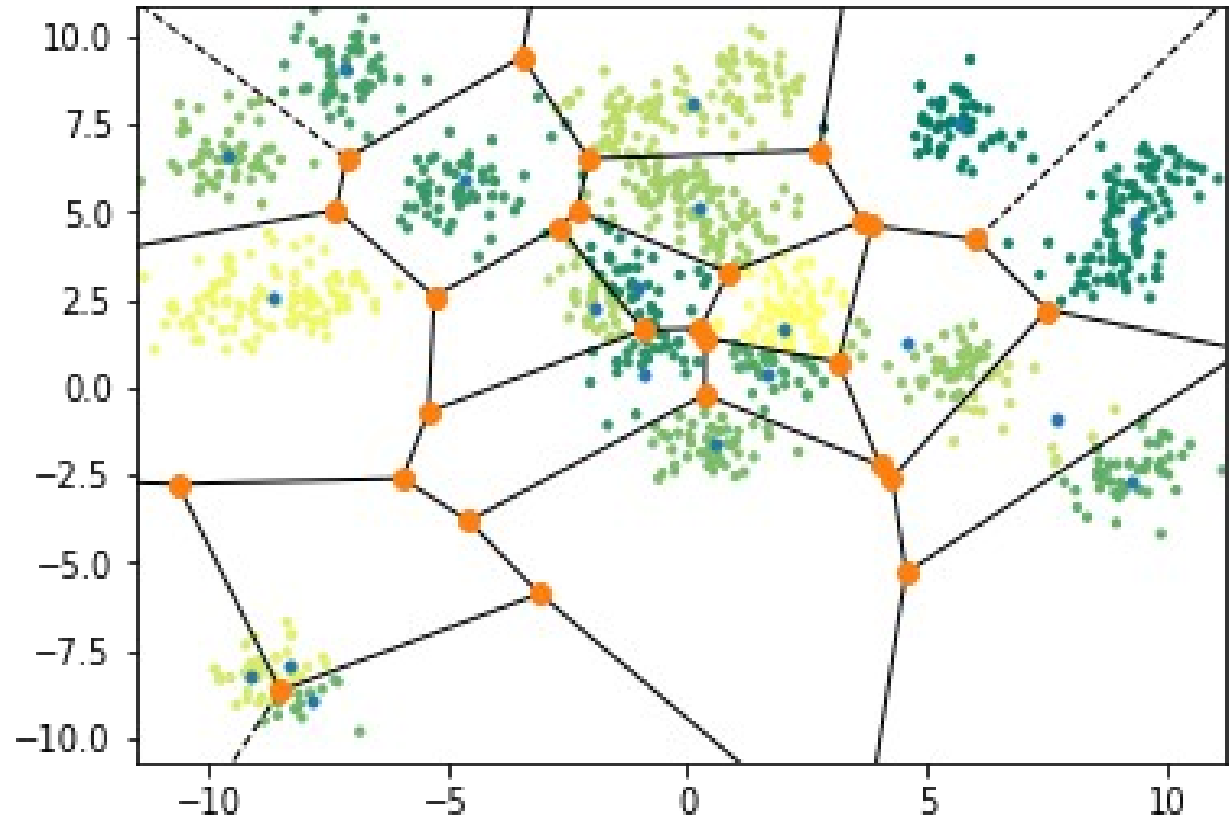
Compositionality



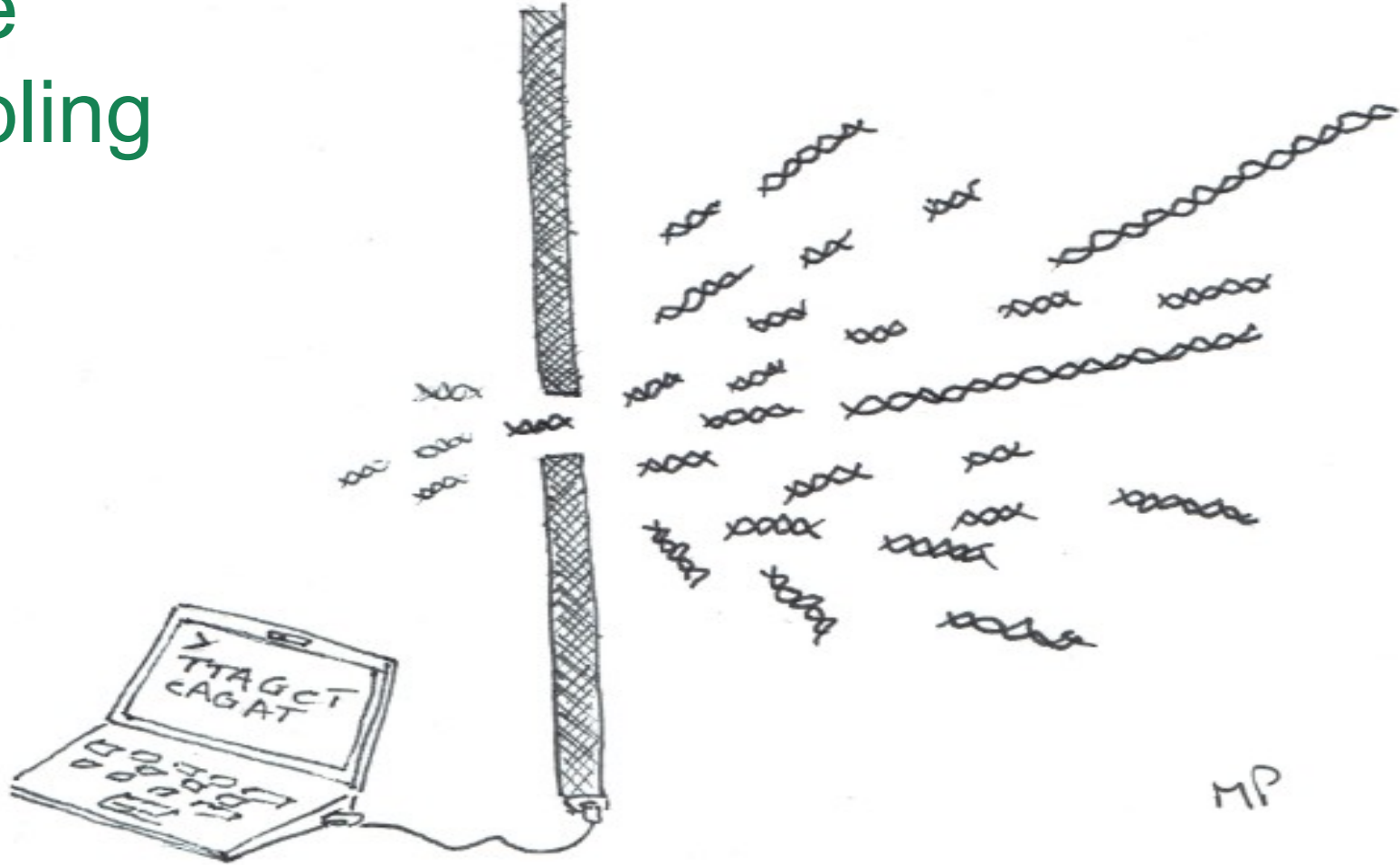
Vandeputte et al. Nature 551:507-511, 2017

Non-parametric clustering: Voronoi regions

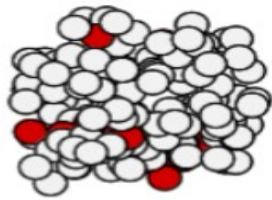
The task:
find *centroids that
describe the data*



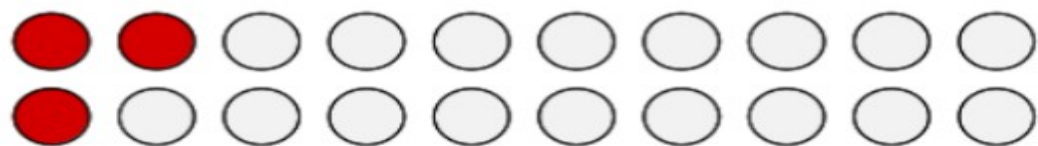
Finite sampling



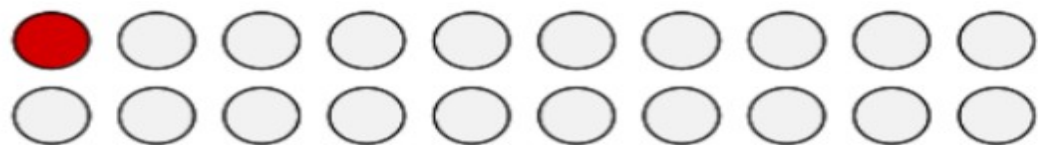
The Poisson distribution



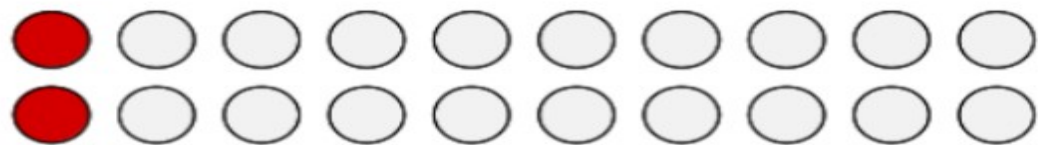
- This bag contains very many small balls, 10% of which are red.
- Several experimenters are tasked with determining the percentage of red balls.
- Each of them is permitted to draw 20 balls out of the bag, without looking.



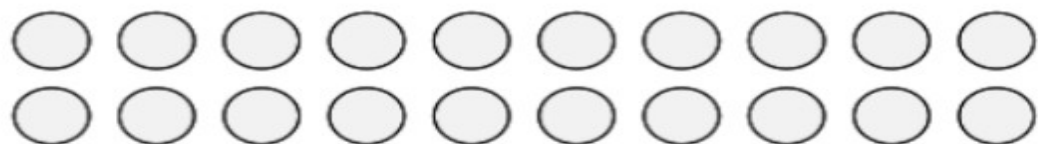
$$3 / 20 = 15\%$$



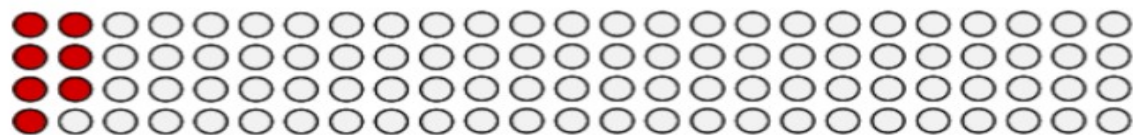
$$1 / 20 = 5\%$$



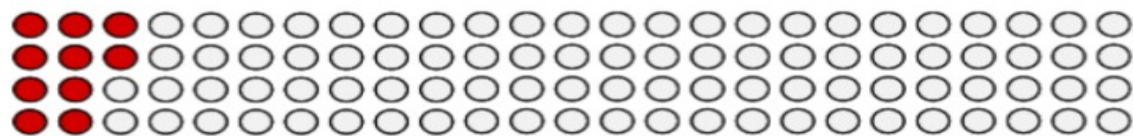
$$2 / 20 = 10\%$$



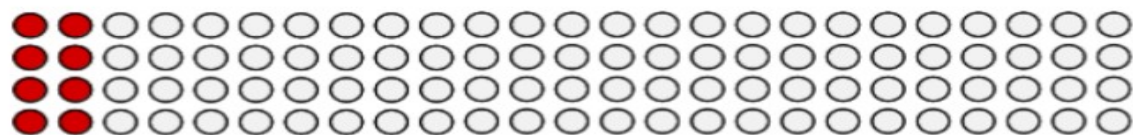
$$0 / 20 = 0\%$$



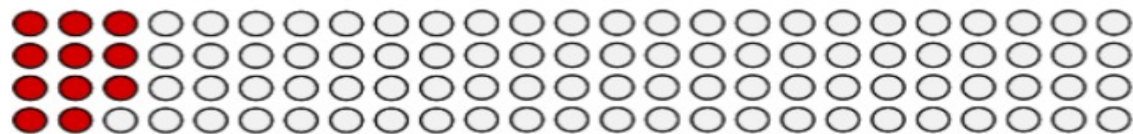
$$7 / 100 = 7\%$$



$$10 / 100 = 10\%$$



$$8 / 100 = 8\%$$



$$11 / 100 = 11\%$$

Poisson distribution: Counting uncertainty

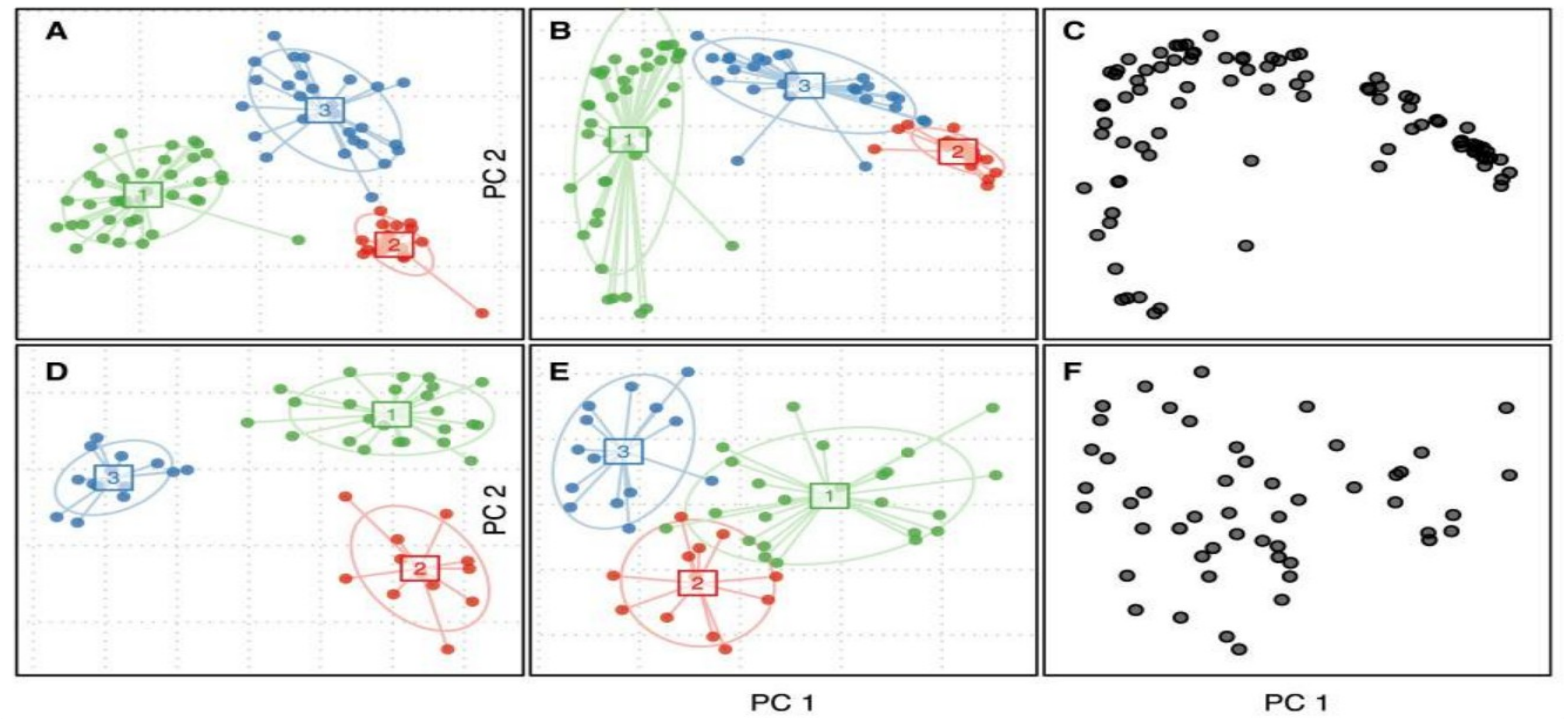
expected number of red balls	standard deviation of number of red balls	relative error in estimate for the fraction of red balls
10	$\sqrt{10} = 3$	$1 / \sqrt{10} = 31.6\%$
100	$\sqrt{100} = 10$	$1 / \sqrt{100} = 10.0\%$
1,000	$\sqrt{1,000} = 32$	$1 / \sqrt{1000} = 3.2\%$
10,000	$\sqrt{10,000} = 100$	$1 / \sqrt{10000} = 1.0\%$

Distinct clusters or extremes on a continuum?

Common Visualizations Can Support Different Conclusions

Soil
samples
with varying
pH

Simulated
data with
no cluster
structure

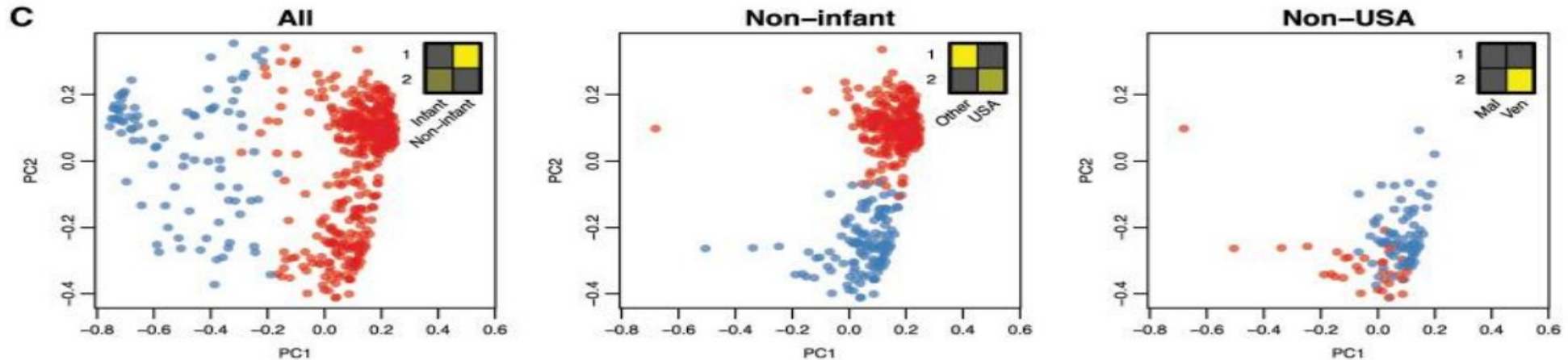


Supervised

Unsupervised
with colors

Unsupervised
without colors

External covariates can induce distinct clusters



Rethinking “Enterotypes”

Dan Knights • Tonya L. Ward • Christopher E. McKinlay • ... Antonio Gonzalez • Daniel McDonald • Rob Knight

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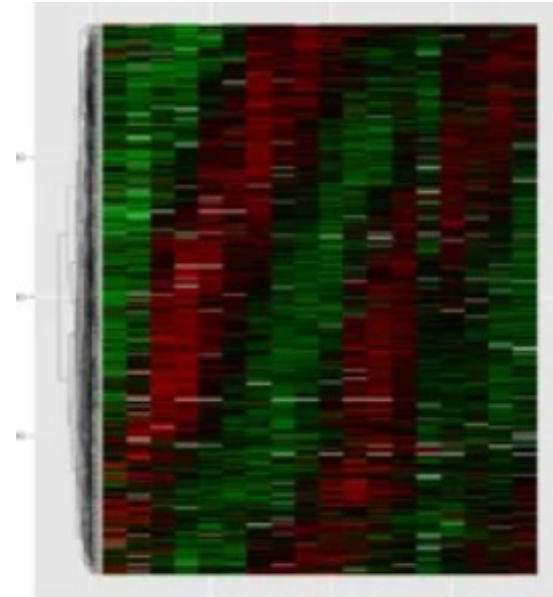
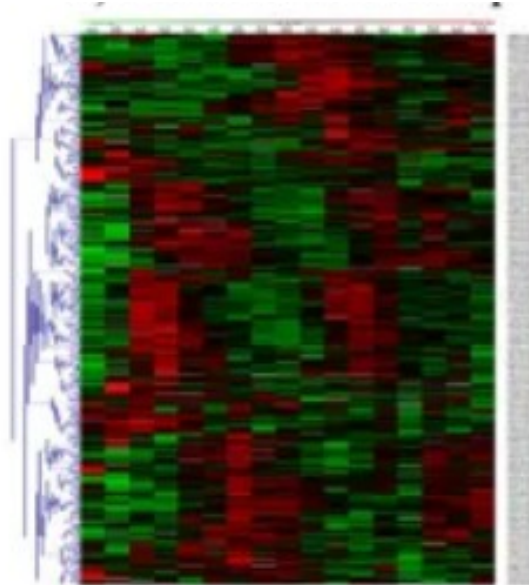
[Open Archive](#) • DOI: <https://doi.org/10.1016/j.chom.2014.09.013> • [Check for updates](#)

NeatMap - non-clustering heat map alternatives in R

[Satwik Rajaram](#)  & [Yoshi Oono](#)

[BMC Bioinformatics](#) 11, Article number: 45 (2010) | [Cite this article](#)

26k Accesses | 47 Citations | 2 Altmetric | [Metrics](#)



Typical study designs

Case-control studies

Interventions

Cross-sectional population cohorts

Prospective follow-ups

Longitudinal time series

Multi-omics

Take-home messages

observations are always incomplete and noisy

generative models describe how the data is (presumably) generated

probabilistic programming can support applications

Latent variable modeling for the microbiome FREE

Kris Sankaran ✉, Susan P Holmes

Biostatistics, kxy018, <https://doi.org/10.1093/biostatistics/kxy018>

Published: 03 June 2018 Article history ▼

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SUMMARY

The human microbiome is a complex ecological system, and describing its structure and function under different environmental conditions is important from both basic scientific and medical perspectives. Viewed through a biostatistical lens, many microbiome analysis goals can be formulated as latent variable modeling problems. However, although probabilistic latent variable models are a cornerstone of modern unsupervised learning, they are rarely applied in the context of microbiome data analysis, in spite of the evolutionary, temporal, and count structure that could be directly incorporated through such models. We explore the application of probabilistic latent variable models to microbiome data, with a focus on Latent Dirichlet allocation, Non-negative matrix factorization, and Dynamic Unigram models. To develop guidelines for when different methods are appropriate, we perform a simulation study. We further illustrate and compare these techniques using the data of Dethlefsen and Relman (2011, Incomplete recovery and individualized responses of the human distal gut microbiota to repeated antibiotic perturbation. *Proceedings of the National Academy of Sciences* 108, 4554–4561), a study on the effects of antibiotics on bacterial community composition. Code and data for all simulations and case studies are available publicly.



Figure 5: The online version provides the text in HTML, data files and up-to-date code.

